

Evaluation Report & Next-Step Execution Roadmap

Intern _1: EVLIN SARA JOHNY

Intern _2: LAKSHMI PRAMODE

Assigned Construct: Emotional Regulation

Proposed Task: Crisis Dispatcher Simulation

Evaluation Status: Proceed with Modifications and Enhanced Behavioral Measurement

1. Executive Summary

The proposed Crisis Dispatcher Simulation demonstrates excellent alignment with our vision of developing an AI-driven, scientifically grounded cognitive and socio-emotional assessment platform. The simulation is highly engaging, adaptive, and capable of capturing behavioral responses under stress. The proposed framework directly targets Emotional Regulation dimensions such as stress tolerance, resilience, motivation, self-regulation, and emotional stability. However, in its current form, task performance may also reflect Processing Speed, Working Memory, gaming experience, and decision-making skills. To align with our startup objectives, the task should focus on measuring how students regulate emotions and recover from setbacks rather than simply how quickly they complete tasks.

2. Alignment with MentiScope's Objectives

- Alignment with Emotional Regulation Framework: 9.5/10
- Scientific Validity: 8.5/10
- Innovation: 9.5/10
- Adaptive Potential: 9/10
- AI Readiness: 9.5/10
- MVP Feasibility: 8.5/10
- Overall Evaluation: 9/10 – Proceed with Modifications

3. Scientific Assessment

Primary emotional regulation components to measure:

- Stress Tolerance
- Emotional Self-Regulation
- Resilience and Recovery
- Motivation Persistence
- Emotional Stability

Potential construct contamination:

- Processing Speed (Gs)
- Working Memory (Gsm)
- Gaming Experience
- Decision-Making Ability

The task should primarily measure how students respond to stress, adapt after failures, and regulate emotions during challenging situations.

4. Recommended Task Architecture

Module 1 – Baseline Scenario

Students handle simple emergency situations with minimal pressure.

Measures: Baseline decision accuracy and emotional stability.

Module 2 – Stress Induction

Introduce time pressure, interruptions, and increasing complexity.

Measures: Stress tolerance and emotional control.

Module 3 – Failure and Recovery

Introduce unexpected setbacks and negative outcomes.

Measures: Recovery time, resilience, and adaptation.

Module 4 – Adaptive Challenge

Present dynamic scenarios with varying stress levels.

Measures: Emotional self-regulation, persistence, and decision stability.

5. Recommended Scoring Model

- Stress Tolerance Score – 25%
- Recovery and Resilience Score – 30%
- Decision Stability Score – 20%
- Adaptation and Persistence Score – 25%

6. AI Analytics to Capture

- Stress recovery time
- Panic clicking frequency
- Error escalation patterns
- Persistence score
- Decision stability index
- Adaptation rate

- Accuracy after setbacks
- Strategy changes
- Emotional recovery curve
- Behavioral consistency under pressure

7. MVP Deliverables (Current Phase)

- ✓ Scenario event generator
- ✓ Stress trigger library
- ✓ Recovery analytics engine
- ✓ Emotional regulation scorecard
- ✓ Event logger
- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data:

student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data:

event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output:

raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output:

behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

Sprint Objective:

Redesign the simulation to prioritize emotional regulation indicators and develop the foundational MVP prototype.

Expected Deliverables:

1. Literature review on Emotional Regulation and Resilience frameworks.
2. Construct map defining stress tolerance, resilience, self-regulation, and emotional stability.
3. Emergency scenario library with multiple stress levels.
4. Stress trigger specifications (time pressure, interruptions, failures).
5. Recovery analytics specifications.
6. Event logging implementation.
7. Emotional scorecard design.
8. Demonstration prototype of one simulation scenario.

10. Definition of Done (DoD)

Scientific:

- ✓ Emotional regulation components clearly defined.
- ✓ Construct contamination minimized.
- ✓ Stress and recovery indicators validated.

Technical:

- ✓ Event generator implemented.
- ✓ Stress trigger library developed.
- ✓ Recovery analytics engine functional.
- ✓ Event logging and analytics dashboard available.
- ✓ Results exported in SDK format.

Validation:

- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with at least 100 students.
- ✓ Preliminary evidence of emotional regulation measurement.
- ✓ Correlation studies with Attention and Working Memory.

Guidance to the Interns

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Emotional Regulation that follows the common Assessment SDK. Focus on capturing behavioral responses to stress, resilience, recovery patterns, and emotional self-regulation. This module will later be integrated into the unified cognitive and career assessment platform.